Data Engineering & Automation Intern: Technical Exercise

Before you start

Please spend no more than 1 hour on this exercise. We care far more about how you approach the problem than whether you finish, and an unfinished solution with clear thinking is a great result.

You can use any tools you normally would, including Google, LLMs and AI assistants. Just make sure you understand and can explain everything you submit, because we will talk through it together in the interview.

Background

Many UK homes have cavity walls: two layers of brick with a gap between them, which may or may not be filled with insulation. To check, surveyors drill a small hole and look inside with a camera called a borescope, then write up what they found in a borescope report.

These reports are a valuable source of ground truth for validating our models. The problem is that every surveyor writes them differently, so the same finding can be described in many ways. Your job in this role would be turning reports like these into clean, structured data, and this exercise is a small taste of that.

The task

Below are three text snippets taken from borescope reports written by different surveyors. Write a Python function that takes one snippet and returns a structured record describing the cavity fill status.

Designing the output structure is an important part of this exercise, and you decide what it looks like. At a minimum it should tell us whether the cavity is fully filled, partially filled, empty, or something else. Think about what someone querying this data across thousands of buildings would need, and add any fields you think are useful.

You can use this stub, or structure it differently if you prefer:

```
def parse_borescope(text: str):
    """Parse a borescope report snippet."""
    ...
```

We will also be looking at:

- The structure of your output and why you chose it
- How you handle wording that differs between surveyors
- What your function does when a report is unclear or incomplete
- How you check that your function works
- Any decisions or trade-offs you made

The snippets

Snippet A

```
Property: 14 Elm Road, Leeds, LS6 2AB
Inspection date: 03/06/2025
Wall type: Masonry cavity
Cavity width: 50mm
Insulation present: Yes
Insulation type: Blown mineral fibre
Fill: Full
```

Snippet B

```
Property: Flat 2, 8 Victoria Street, York, YO1 6GF

Borescope inspection carried out at the front and rear elevations.
The cavity measured approximately 60-65mm. Polystyrene bead
insulation was observed but the fill was inconsistent, with an
estimated 40% coverage at the rear and voids visible near the
first floor joist line. Recommend top-up.
```

Snippet C

```
Property: 31 Mill Lane, Bradford, BD5 0HN
Drill point 1 (front, ground floor): cavity 75mm, no insulation
Drill point 2 (side, first floor): cavity 75mm, no insulation
Drill point 3 (rear, ground floor): cavity 70mm, debris present,
unable to confirm insulation
Overall: Cavity appears unfilled. CWI not present.
```

Glossary

- **CWI**: cavity wall insulation
- **Common insulation types**: blown mineral fibre, polystyrene (EPS) beads and polyurethane foam are all materials used to fill cavities
- **Voids**: gaps in the cavity where insulation should be but is not
- **Debris**: rubble or mortar sitting in the cavity, which can block the view or the insulation
- **Elevation**: one side of the building, such as the front, rear or side
- **Joist line**: the level where the floor beams meet the wall, usually between storeys
- **Top-up**: adding more insulation to a cavity that is only partly filled

What to send us

Please email your solution at least 24 hours before your interview. Ideally, share it as a link to a GitHub repository. If that is not possible, a single Python file or a small zipped folder is fine.

Include a few lines on the decisions you made and anything you would do differently with more time. Bullet points are perfectly fine.

The interview

The interview is 30 minutes, remote over Google Meet. Please join from a computer with your solution open, as you will be sharing your screen.

1. You present your solution and walk us through what you have done (about 5 minutes)
2. We ask questions about your solution and the choices you made (about 15 minutes)
3. A hypothetical discussion about how you would extend your solution and fit it into a wider data pipeline (5 to 10 minutes)

Remember, we want you to do well. This is not about catching you out, and we are looking forward to seeing how you think.